

Weight-Space Posterior

The explicit feature representation of the prior admits a complementary weight-space formulation of Gaussian process inference. Rather than conditioning the process directly through its covariance kernel, we infer the latent coefficient vector $\mathbf{w}$ in the expansion

$$E(\mathbf{x}) = \Phi(\mathbf{x})^H \mathbf{w},$$

where $\mathbf{w} \sim \mathcal{N}(0, \mathbf{W})$. The resulting posterior process is identical to that obtained in function space, but the computational cost depends on the number of features rather than the number of observations.

Evaluating the feature functions at the observation points yields the design matrix

$$\Phi = \Phi(X_b) \in \mathbb{C}^{F \times N_b},$$

so that the observation model becomes

$$\mathbf{h} = \Phi^H \mathbf{w} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(0, \sigma_n^2 \mathbf{I}).$$

This is a standard Bayesian linear regression problem. The posterior over the weights is Gaussian,

$$\mathbf{w} \mid \mathbf{h} \sim \mathcal{N}(\mathbf{w}_*, \mathbf{A}^{-1}),$$

with posterior precision

$$\mathbf{A} = \mathbf{W}^{-1} + \frac{1}{\sigma_n^2} \Phi \Phi^H,$$

and posterior mean

$$\mathbf{w}_* = \frac{1}{\sigma_n^2} \mathbf{A}^{-1} \Phi \mathbf{h}.$$

The posterior mean field is obtained by mapping the inferred coefficients back through the feature basis,

$$E_*(\mathbf{x}) = \Phi(\mathbf{x})^H \mathbf{w}_*.$$

The corresponding kernel under the same quadrature rule is

$$\mathbf{K}(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x})^H \mathbf{W} \Phi(\mathbf{y}),$$

which is the finite-dimensional approximation used in the function-space formulation. Substituting this expression into the standard Gaussian process posterior and applying the Woodbury matrix identity yields exactly the weight-space posterior above. Thus, the function-space and weight-space formulations produce identical posterior means and covariances.

The practical distinction is computational. Function-space inference requires inversion of the $N_b \times N_b$ Gram matrix, whereas the weight-space formulation requires inversion of the $F \times F$ matrix $\mathbf{A}$. Since the explicit-polarization construction employs $F = 2N_s$ features, the weight-space formulation is advantageous whenever $F < N_b$, reducing the computational cost from one governed by the number of observations to one governed by the number of features.